Figure 5.6 shows feedback bias of a transistor.

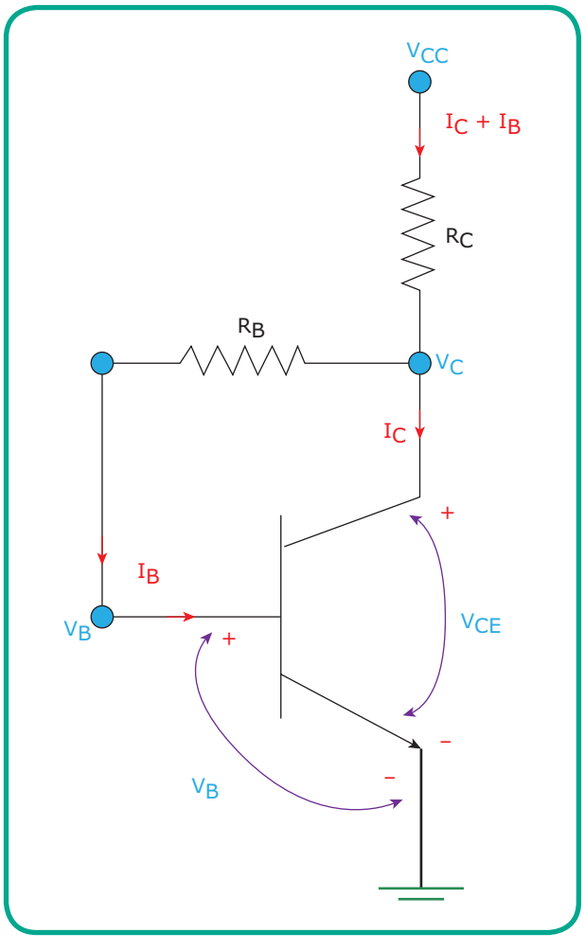


Figure 5.6 Feedback Bias

5.6.3 Voltage Divider Bias

This is the most commonly used biasing arrangement, as it provides good bias stability. The emitter resistance R_E provides stabilization. The resistance R_E causes a voltage drop in a direction so as to reverse bias the emitter junction. Since the emitter base junction has to be forward biased, the base voltage is obtained from the supply through R_1-R_2 network. The net forward bias across the emitter base junction is equal to V_B–DC voltage drop across R_E. The DC bias circuit is independent of transistors β to avoid the loss of AC signal. A capacitor having large capacitance is connected across R_E to bypass the unwanted AC fluctuations. Figure 5.7 shows voltage divider method of a transistor.

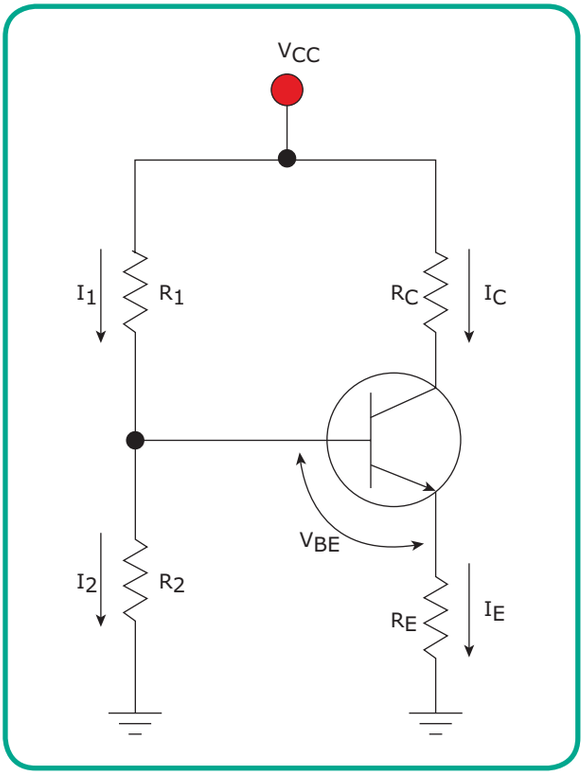


Figure 5.7 Voltage divider bias

5.7 WORKING OF NPN & PNP TRANSISTOR

5.7.1 Working of NPN transistor

Figure 5.8 shows the NPN transistor with forward bias to the emitter-base junction and reverse bias to collector-base junction. The forward bias causes the electrons in the n- type emitter to flow towards the base. This constitute the emitter current I_E. As these electrons flow through the p-type base, they tend to combine with holes. As the base is lightly doped and very thin, only a few electrons (less than 5%) combine with holes to constitute base current I_B. The reminder electrons (more than 95%) cross-over into the collector

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